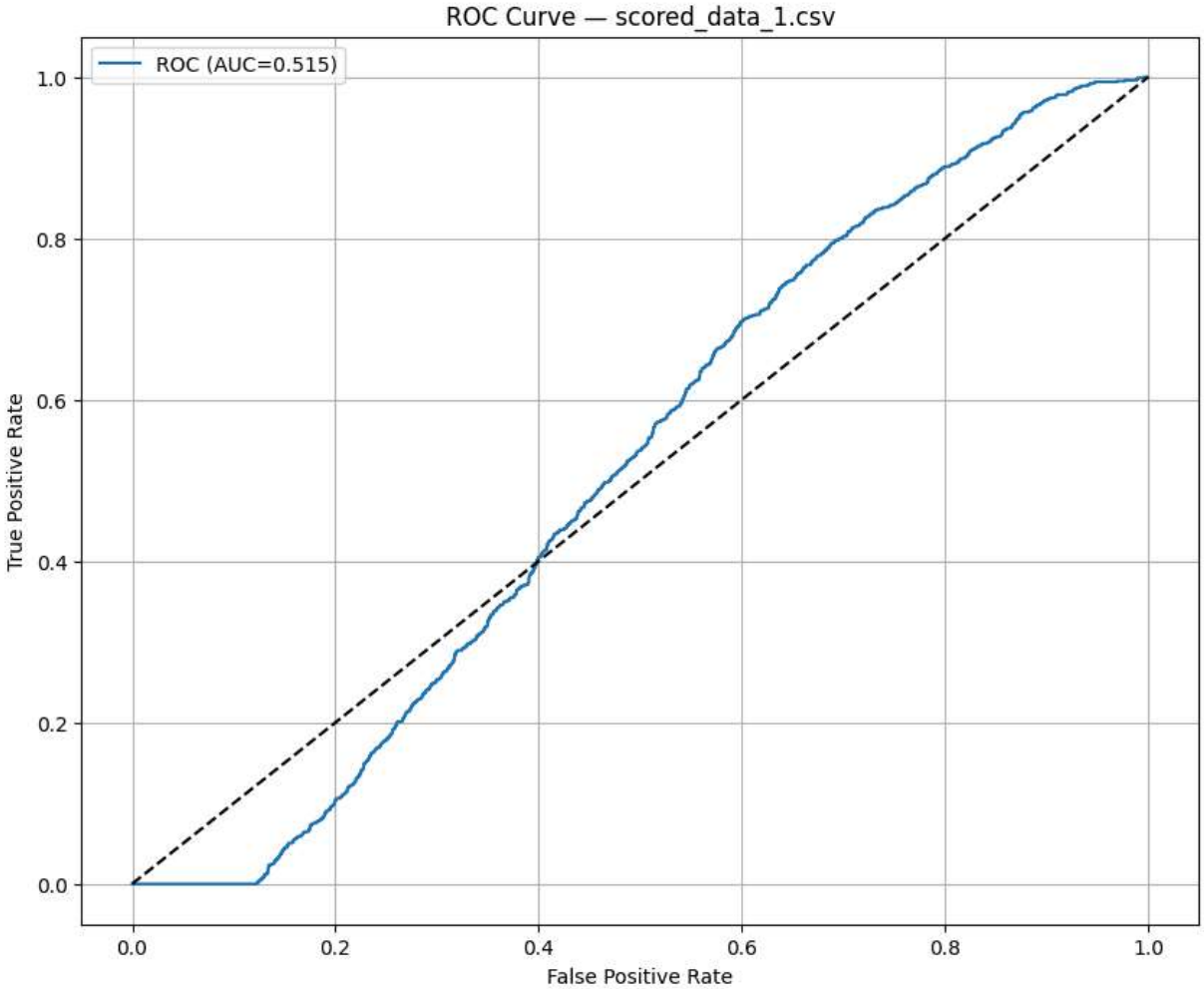
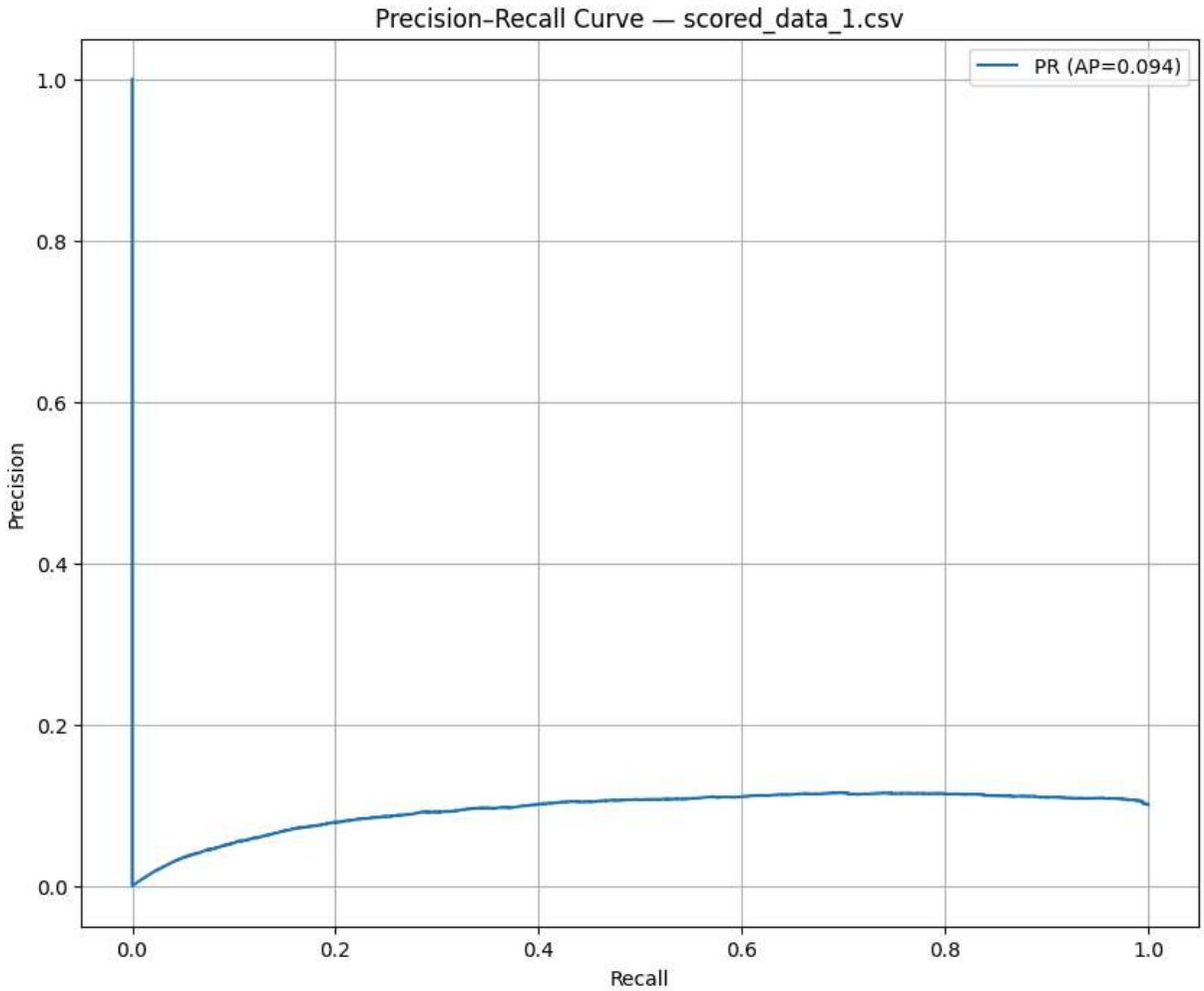
In [5]:
```
# t and generate ROC + PR curves
# ----------------------------------------------------------
for file in files:
    print(f"\nProcessing: {file}")

    # Load dataset
    df = pd.read_csv(file)

    # True labels - ensure they are integers for classification
    y_true = df["label"].astype(int)  # Convert to integers if they're not already

    # Probability columns
    prob_cols = get_prob_columns(df)
    y_score = df[prob_cols].values

    # Class list - make sure it matches the actual classes in your data
    classes = sorted(y_true.unique())  # Get actual classes from the data

    # Binarize labels for multiclass ROC/PR
    y_true_bin = label_binarize(y_true, classes=classes)

    # Check dimensions and adjust if necessary
    if y_score.shape[1] == 1 and len(classes) > 1:
        # If we only have one probability column but multiple classes,
        # we're likely dealing with a binary classification problem
        # where the second class probability is 1-first_class_prob
        y_score = np.column_stack((1-y_score, y_score))

    # Make sure y_score has the right shape for multiclass
    if len(classes) > 2 and y_score.shape[1] != len(classes):
        print(f"Warning: Number of probability columns ({y_score.shape[1]}) doesn't
        continue  # Skip this file if dimensions don't match

    # ----------------------------------------------------------
    # ROC Curve
    # ----------------------------------------------------------
    plt.figure(figsize=(10, 8))

    # Handle binary classification case specially
    if len(classes) == 2:
        # For binary classification, we only need one curve
        fpr, tpr, _ = roc_curve(y_true, y_score[:, 1] if y_score.shape[1] > 1 else
        roc_auc = auc(fpr, tpr)
        plt.plot(fpr, tpr, label=f"ROC (AUC={roc_auc:.3f})")
    else:
        # For multiclass, plot curve for each class
        for i in range(len(classes)):
            fpr, tpr, _ = roc_curve(y_true_bin[:, i], y_score[:, i])
            roc_auc = auc(fpr, tpr)
            plt.plot(fpr, tpr, label=f"Class {classes[i]} (AUC={roc_auc:.3f})")

    plt.plot([0, 1], [0, 1], "k--")
    plt.title(f"ROC Curve — {os.path.basename(file)}")
    plt.xlabel("False Positive Rate")
```

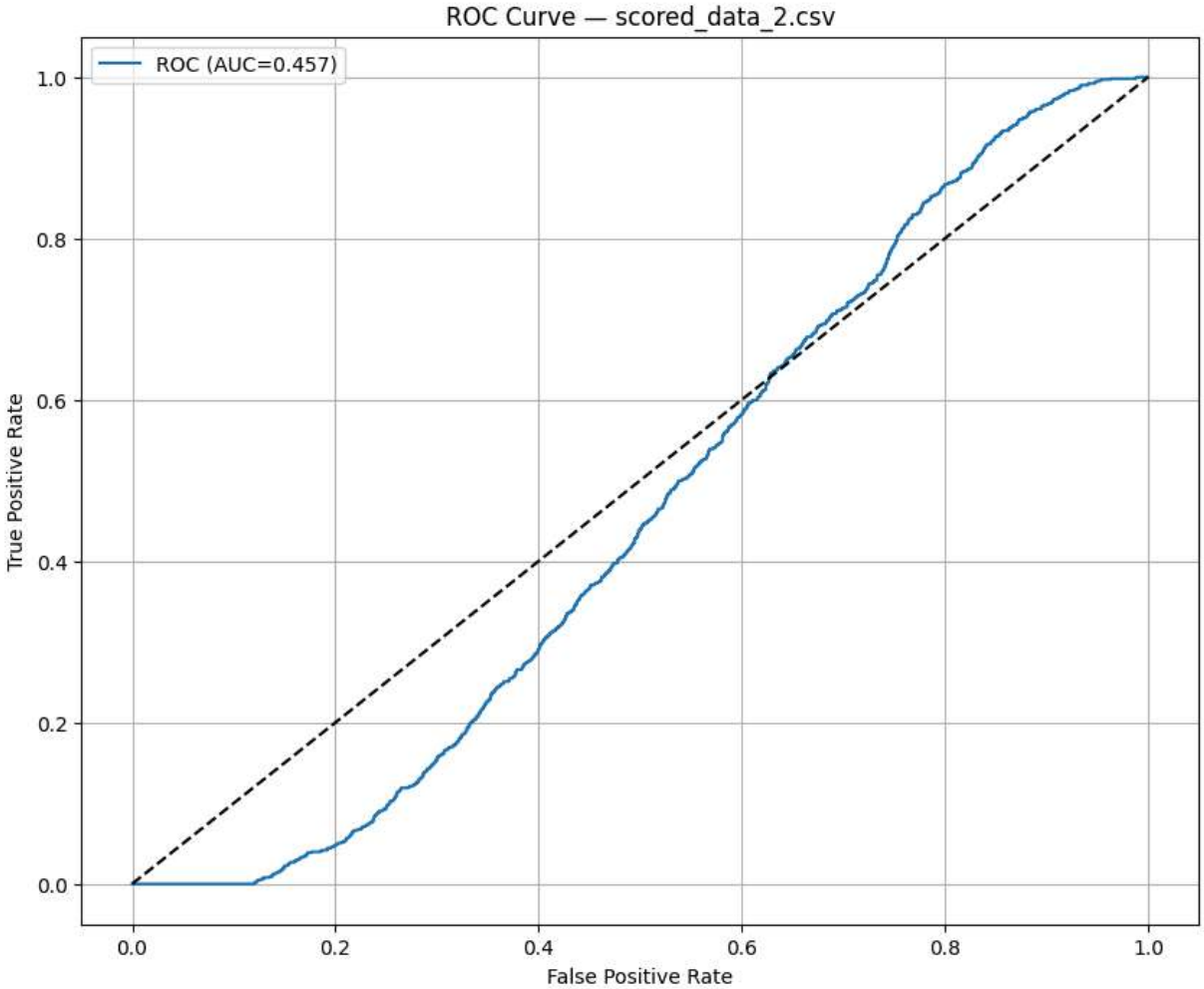
```
    plt.ylabel("True Positive Rate")
    plt.legend()
    plt.grid(True)
    plt.show()

    # ---------------------------------------------------------
    # Precision-Recall Curve
    # ---------------------------------------------------------
    plt.figure(figsize=(10, 8))

    # Handle binary classification case specially
    if len(classes) == 2:
        # For binary classification, we only need one curve
        precision, recall, _ = precision_recall_curve(y_true, y_score[:, 1] if y_sc
        ap = average_precision_score(y_true, y_score[:, 1] if y_score.shape[1] > 1
        plt.plot(recall, precision, label=f"PR (AP={ap:.3f})")
    else:
        # For multiclass, plot curve for each class
        for i in range(len(classes)):
            precision, recall, _ = precision_recall_curve(y_true_bin[:, i], y_score
            ap = average_precision_score(y_true_bin[:, i], y_score[:, i])
            plt.plot(recall, precision, label=f"Class {classes[i]} (AP={ap:.3f})")

    plt.title(f"Precision-Recall Curve - {os.path.basename(file)}")
    plt.xlabel("Recall")
    plt.ylabel("Precision")
    plt.legend()
    plt.grid(True)
    plt.show()
```

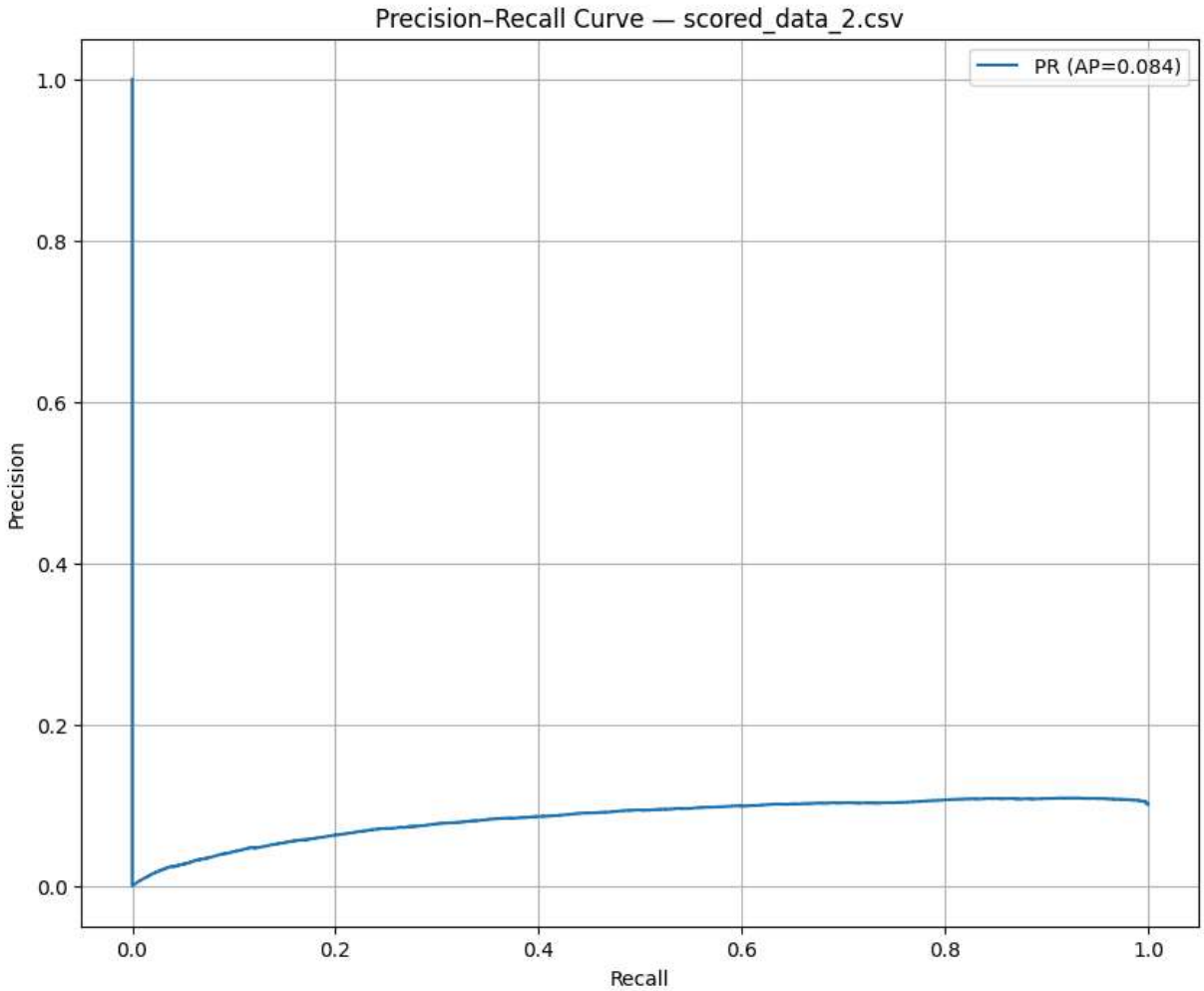
```
Processing: scored_data_1.csv
```


ROC Curve — scored_data_1.csv
ROC (AUC=0.515)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

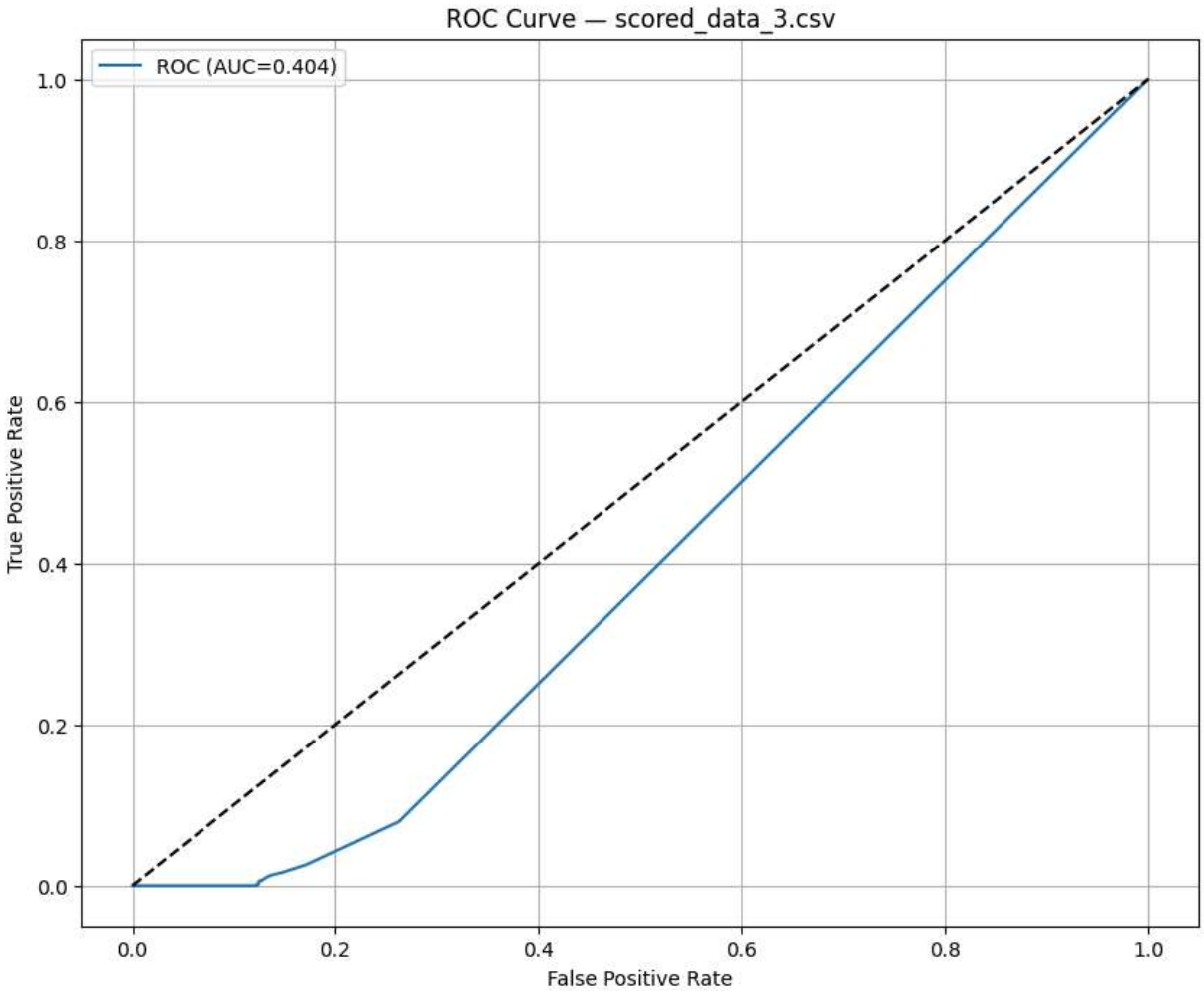


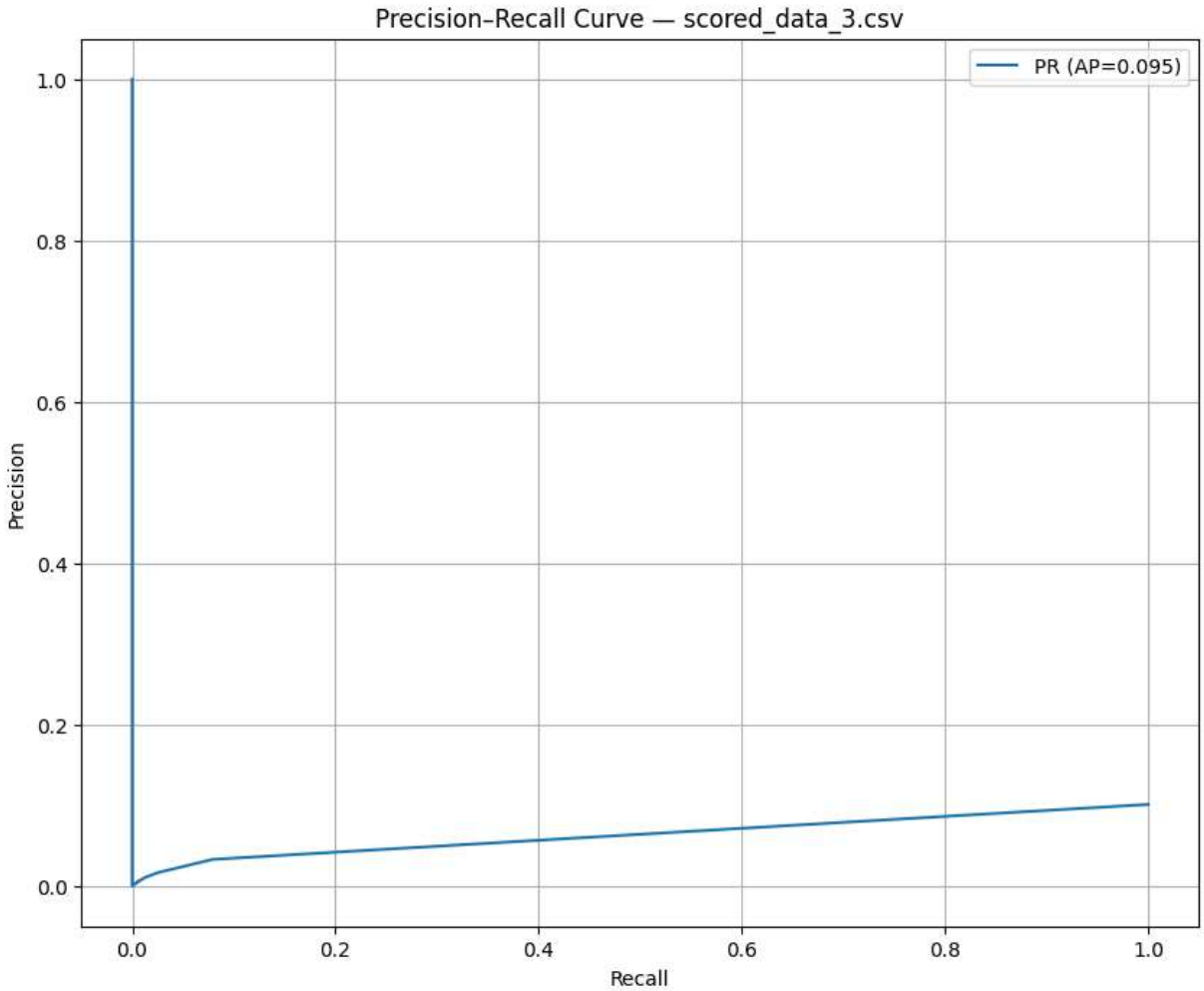

```
Processing: scored_data_2.csv
```


ROC Curve — scored_data_2.csv
ROC (AUC=0.457)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

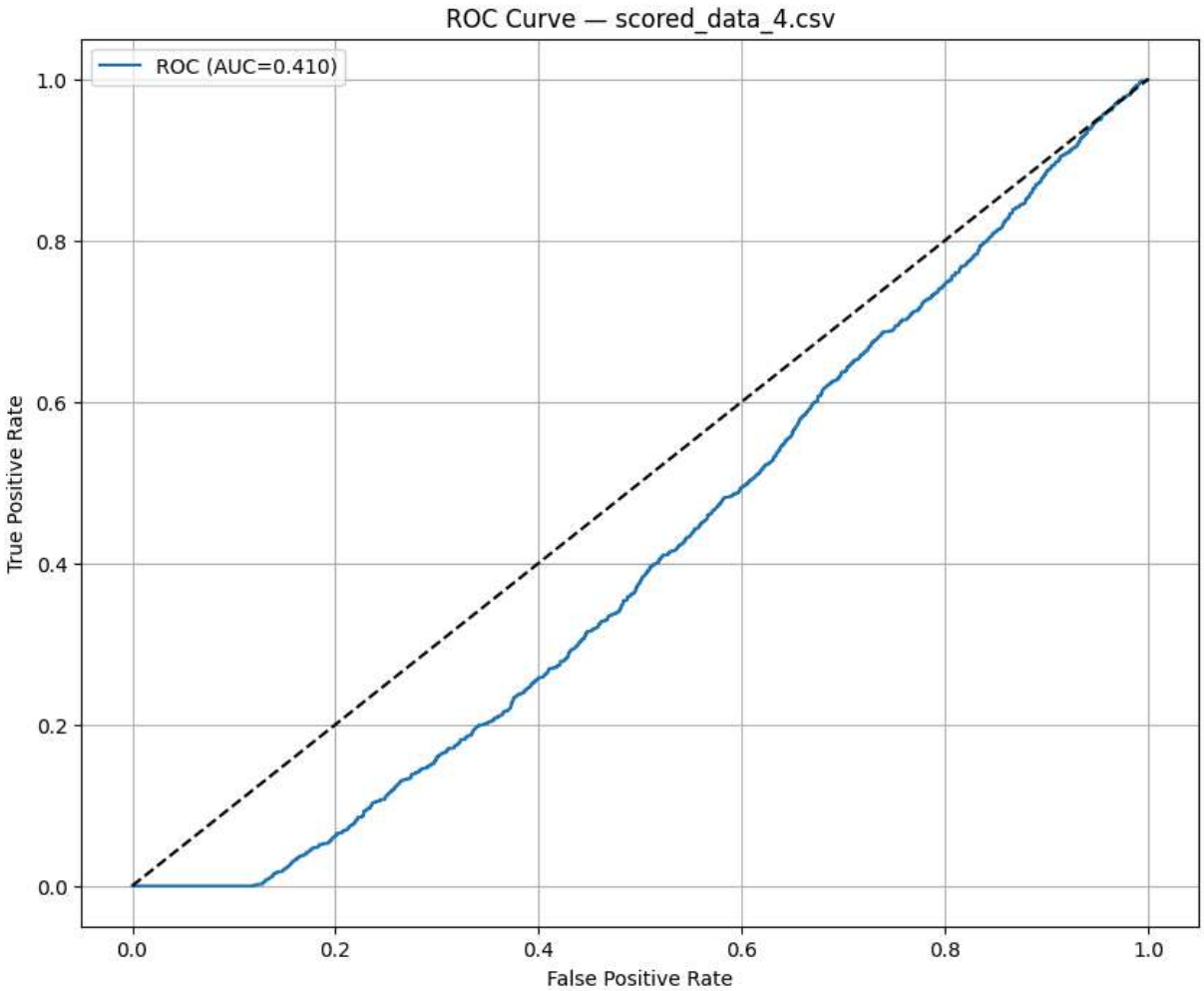


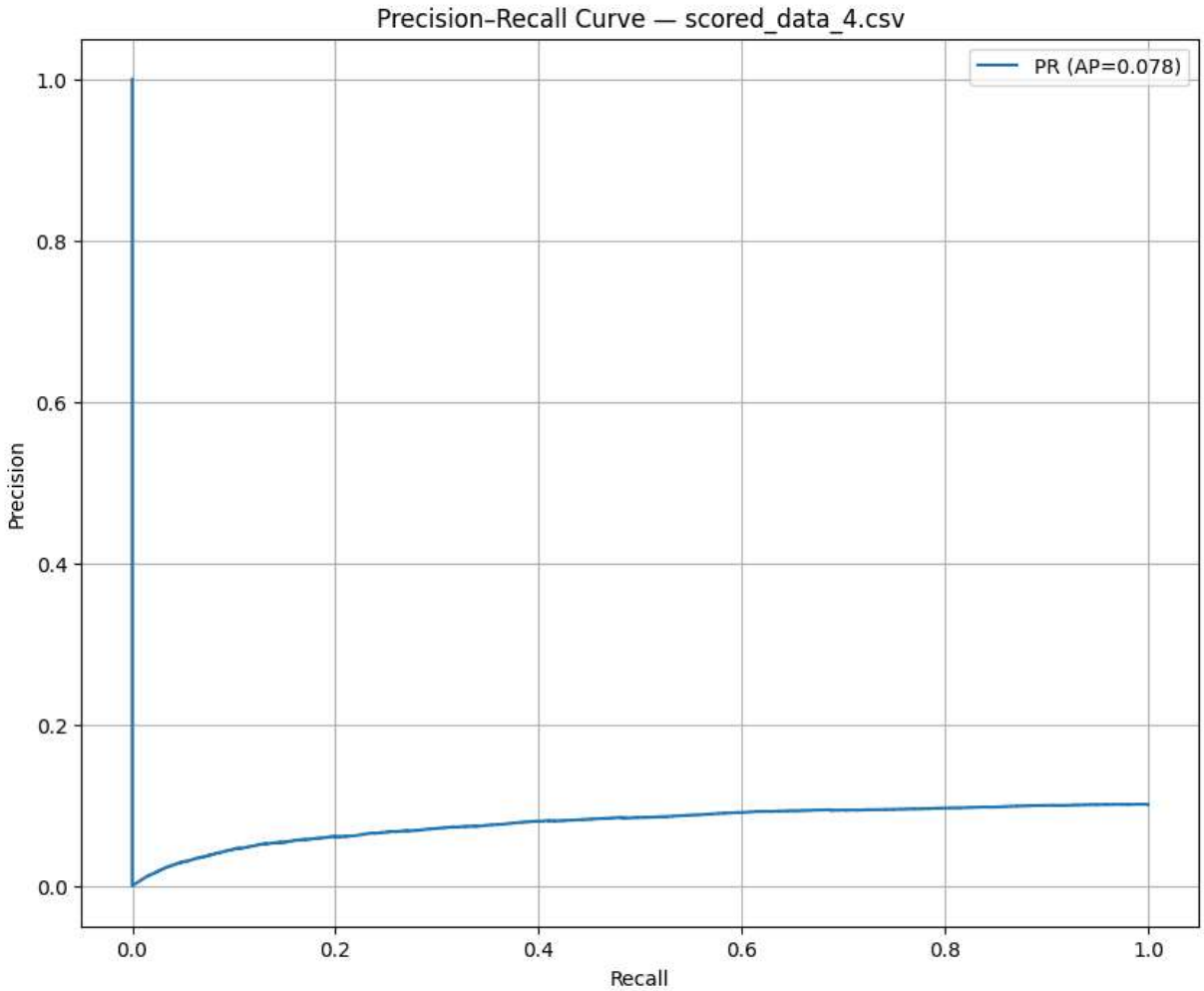

```
Processing: scored_data_3.csv
```


ROC Curve — scored_data_3.csv
ROC (AUC=0.404)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

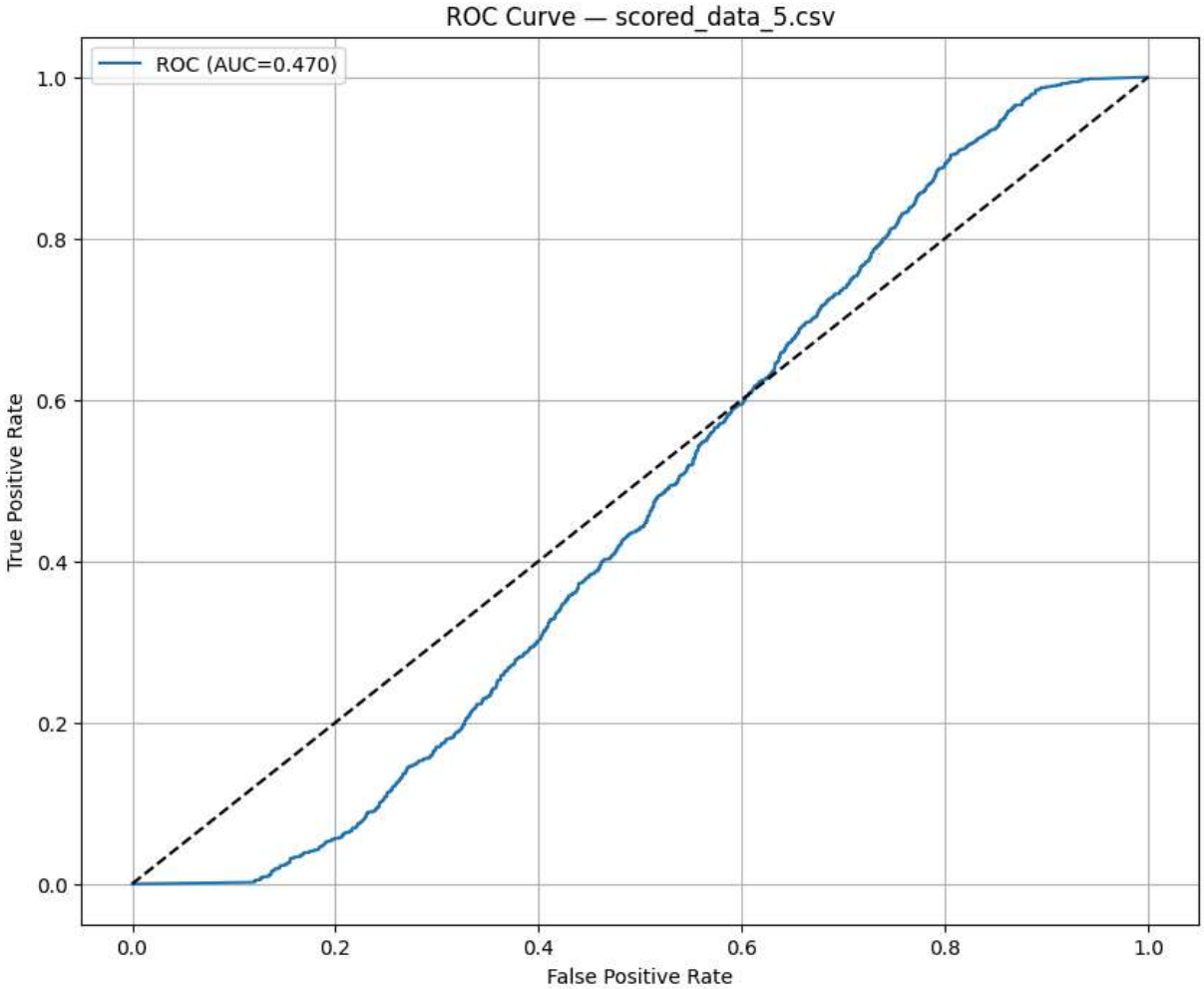


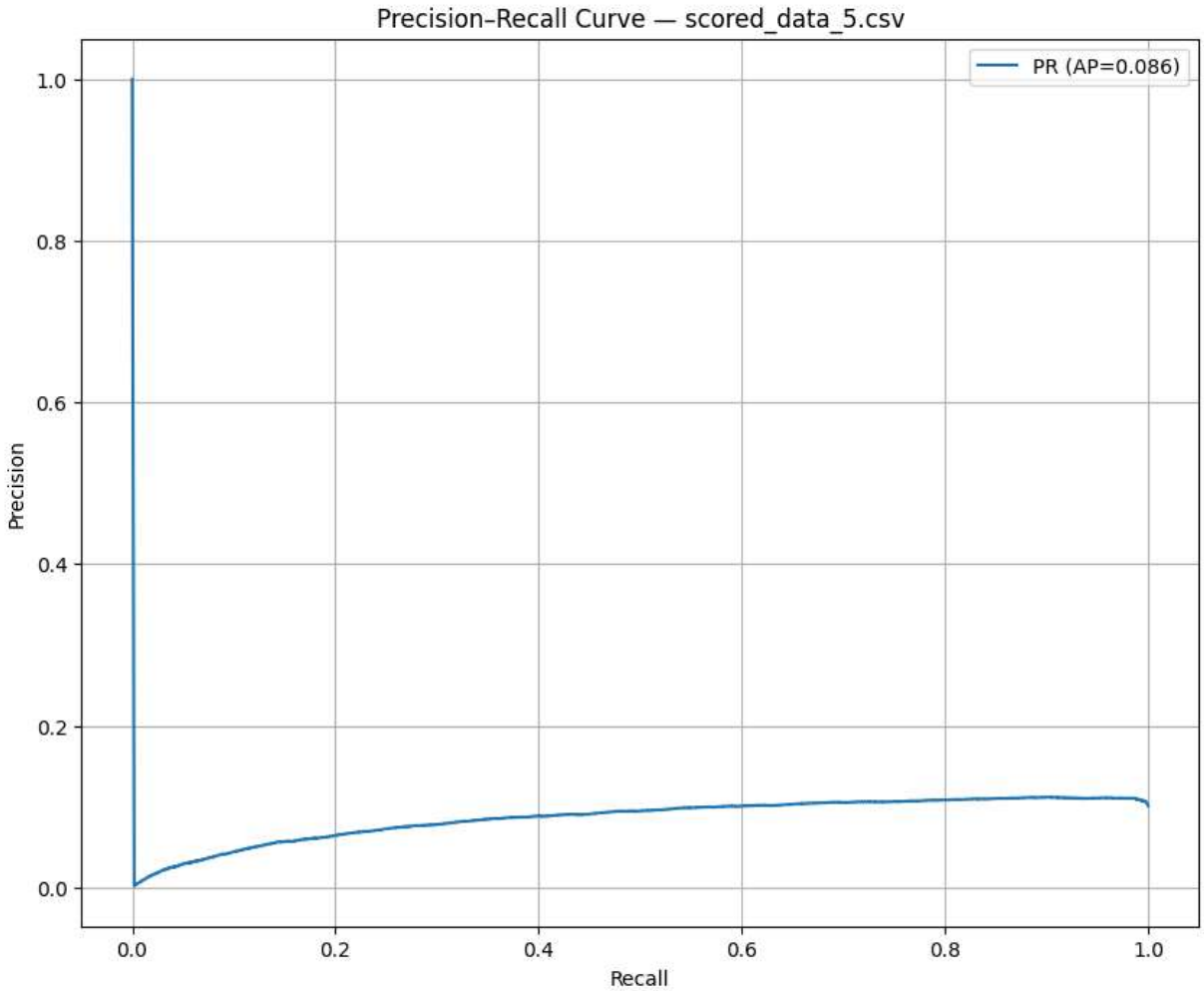

```
Processing: scored_data_4.csv
```


ROC Curve — scored_data_4.csv
ROC (AUC=0.410)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

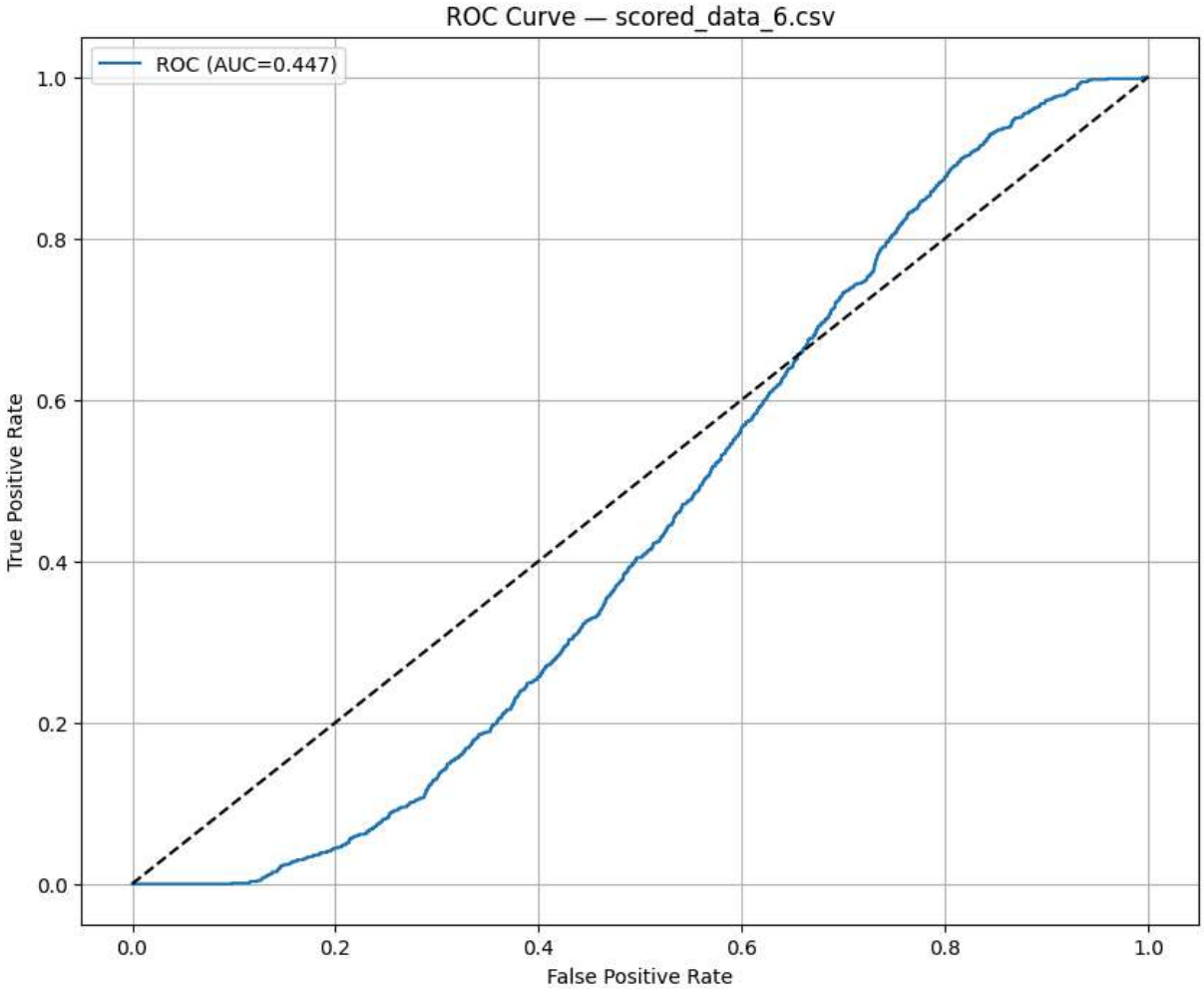


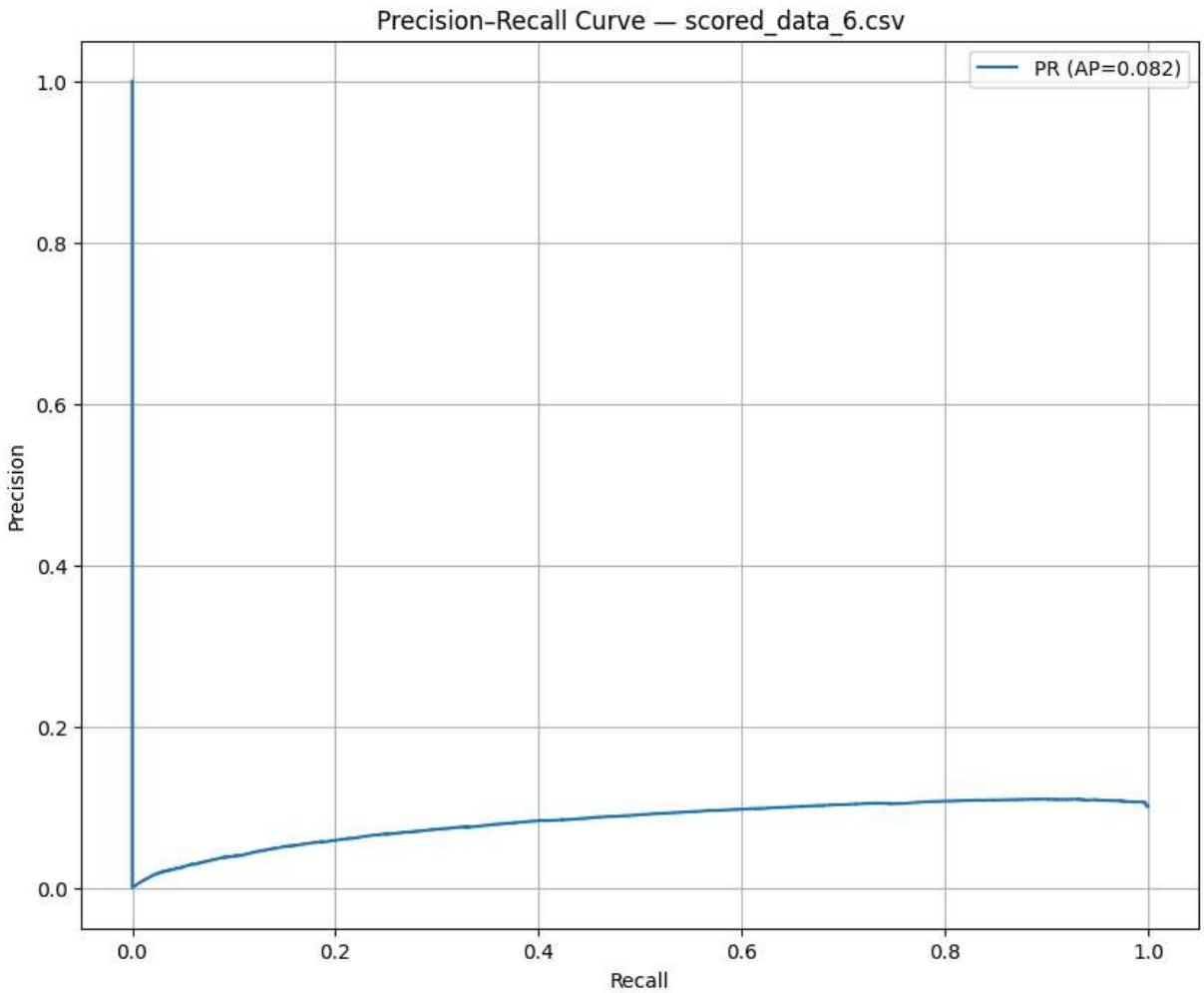

```
Processing: scored_data_5.csv
```


ROC Curve — scored_data_5.csv
ROC (AUC=0.470)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0




```
Processing: scored_data_6.csv
```


ROC Curve — scored_data_6.csv
ROC (AUC=0.447)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0


Precision–Recall Curve — scored_data_6.csv
PR (AP=0.082)
Precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0

In [ ]: